Kuardio × QardioArm: interoperability test report

Version 1.0 · 22 October 2026 · Uladzimir Kazakevich, Kuardio (contact@kuardio.com) · for App Review, guideline 1.4.1

Summary. Kuardio does not measure blood pressure. The QardioArm cuff (FDA 510(k) K140067, CE 2460) measures and computes the values, as it did with the manufacturer's app; Kuardio starts the measurement, receives the result over the standard Bluetooth SIG Blood Pressure Service, shows it and stores it in Apple Health unchanged. This report shows that the transfer works as described: every packet the cuff sends is decoded correctly, a completed reading is reported once with the cuff's values, and error and cancel cases end the reading without saving anything.

1. System under test

App	Kuardio 1.0 (build 5), iOS 26 or later, iPhone. Core Bluetooth and HealthKit only; no server, no third-party code.
Cuff	QardioArm model A100 (Qardio, Inc.), hardware Cp_A2_p2, firmware 1.6, battery 90 % at the start of the session. Non-invasive oscillometric upper-arm monitor: FDA 510(k) K140067 (Class II, 21 CFR 870.1130, DXN); EU Class IIa, CE 2460.
Phone	iPhone 18 Pro Max, iOS 27.0.1 (owner's device; the same one as in the demo video).
Capture tool	Kuardio's built-in Bluetooth log (Settings → About → tap the version five times), which records every discovered service and characteristic, every command written and every packet received, with raw bytes and the parsed values.

2. How Kuardio uses the cuff

- **Standard profile.** The cuff exposes the Bluetooth SIG Blood Pressure Service (0x1810) with the Blood Pressure Measurement characteristic (0x2A35), the Battery Service (0x180F) and Device Information (0x180A). Kuardio reads 0x2A35 and the battery level, and the model, firmware and manufacturer strings from Device Information.
- **Start and stop.** The cuff has no button for the app flow: a measurement is started by writing `F1 01` to the cuff's control characteristic (`583CB5B3-…`) and cancelled with `F1 02`, exactly as the manufacturer's app did. Kuardio writes nothing else to the cuff and never changes its firmware, calibration or algorithm.
- **Decoding.** 0x2A35 packets follow the Bluetooth Blood Pressure Profile: a flags byte, then systolic, diastolic and mean arterial pressure as IEEE-11073 SFLOAT, optional pulse and measurement status. While the cuff inflates and deflates it sends the current cuff pressure in the systolic field with diastolic and MAP at 0 (flags 0x00); the final result has flags 0x14 (pulse and status present). SFLOAT NaN (`07FF`) in the systolic field is the cuff's error packet.
- **Storage.** The decoded systolic, diastolic and pulse are written to Apple Health as a blood-pressure correlation and a heart-rate sample with the same numbers; nothing is rounded, scaled or computed by Kuardio.

3. Part A: live session with the real cuff

Recorded 28 September 2026, 11:27–11:29 UTC, from the first connection to the last packet. The full log (serial number and system ID removed) is published at docs/ble/captures/2026-09-28-qardioarm-fw1.6.json; the complete packet list is in the appendix.

Run	Start (F1 01)	Packets	Peak cuff pressure	Outcome
1	11:27:51	38	101	Stopped by the cuff at 11:28:05 with an error packet `04 FF 07 01 00 03 00 00 00` (SYS = NaN): shown as an interrupted reading, nothing saved.
2	11:28:13	104	165	Completed at 11:28:45: final packet `14 90 00 5F 00 6F 00 60 00 00 00` → sys=144.0 dia=95.0 map=111.0 pulse=96.0 status=none flags=0x14 unit=mmHg. Sent twice by the cuff; Kuardio reports it once.
3	11:28:52	12	34	Cancelled by the user at 11:28:58 (F1 02). The cuff deflated and sent no further packets.
4	11:29:02	8	22	Stopped by the cuff at 11:29:07 with an error packet `14 FF 07 01 00 03 00 00 00 00 00` (SYS = NaN): shown as an interrupted reading, nothing saved.

Observed during the session: the battery characteristic notified 90 % → 84 % mid-measurement, which Kuardio shows on the Measure screen; the final packet of a completed reading is sent twice by the cuff and reported once by Kuardio; after a cancel the cuff sends nothing more and Kuardio returns to the ready state.

4. Part B: automated verification

The parser and the measurement logic are a Swift package with 90 automated tests (Swift Testing), run before every release. The ones relevant to the hardware are:

- **BLSParser**: SFLOAT decoding including the reserved values (NaN, ±infinity, NRes), the flags byte, mmHg and kPa units, timestamp, pulse, user ID and measurement status, and tolerance of trailing bytes.
- **RealCuffCaptureTests** replay the packets of Part A through the parser and the measurement state machine: every captured packet parses; in-progress packets carry the cuff pressure with diastolic and MAP zero; the final packet decodes to 144/95, MAP 111, pulse 96 with no status flags; a completed reading finishes once although the final packet is sent twice; live pressure follows the capture with a 165 mmHg peak before the final packet; the cuff's error packet ends the reading at once; an error packet while inflating is an interrupted reading; cancelling sends F1 02 and returns to idle.
- **MeasurementMachine and MeasurementSession**: battery check before starting, 1–3 averaged readings with a rest between them, the 1.5 s finalize debounce, the 15 s stall detection and the 180 s watchdog, cancel, and visitor mode (nothing saved).
- **Health**: the correlation and heart-rate samples written for a reading carry exactly the parsed values and the reading's metadata; readings outside the cuff's range are not saved.
- **UI tests** on a simulated cuff that replays the same packet shapes: a single reading is measured and saved, a visitor reading is not saved, and Apple's accessibility audit passes on every screen.

5. Part C: value integrity, end to end

To be completed (procedure in 6.1).

#	Time	Final packet from the cuff	Decoded	Shown on the result screen	Stored in Apple Health	Match
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

6.1 Procedure

1. Fresh batteries in the cuff; Kuardio build 5 on the iPhone; Apple Health access allowed.
2. Open the Bluetooth log (Settings → About → tap the version five times), tap *Scan*, wait for *Connected*.
3. Take ten readings from the Measure tab, at least one minute apart, seated, cuff at heart level. After each, note the values on the result screen.
4. Export the log; open the Health app → Browse → Heart → Blood Pressure and Heart Rate; compare each entry with the final packet in the log and with the result screen.

7. Error handling

Situation	What the cuff does	What Kuardio does
Arm movement, cuff can't measure	Stops and sends one packet with SYS = NaN	Ends the reading at once: "Measurement interrupted", nothing saved, offers to try again
Cuff goes silent	—	After 15 s without packets: "Measurement interrupted"; after 180 s in any case
User cancels	Deflates, sends nothing	Writes F1 02, returns to ready
Bluetooth off, cuff out of range, pairing refused	—	Status card with the cause and the fix; no measurement is started
Battery low / critical	Battery service	Warns before starting / refuses to start
Values outside the cuff's specified range	—	Shown with "Not saved — check values"; not written to Health

8. Limitations

This report covers the transfer of values from the cuff to the app and to Apple Health. The accuracy of the measurement itself is the cuff's: it was clinically validated for the 510(k) clearance (ANSI/AAMI ISO 81060-2:2009, EN 1060-4:2004) and in four peer-reviewed ESH-IP 2010 validation studies listed below. Kuardio makes no accuracy claim of its own, does not interpret readings and gives no medical advice; it is a wellness logging app.

9. References

1. FDA 510(k) K140067, QardioArm A100: clearance letter, summary and indications for use (18 June 2014). Performance and clinical test: ANSI/AAMI ISO 81060-2:2009, EN 1060-4:2004. The summary states the device "requires an external device (e.g. a smartphone) to display results".
2. FDA 510(k) K220106, QardioArm 2 (15 June 2022), predicate K140067.
3. EU Declaration of Conformity, QardioArm A100, Class IIa, MDD 93/42/EEC, notified body 2460 (Qardio Netherlands B.V., 1 November 2018).
4. Chahine MN et al. Validation of BP devices QardioArm in the general population and Omron M6 Comfort in type II diabetic patients according to the ESH International Protocol. *Med Devices (Auckl)*. 2018;11:11–20. doi:10.2147/MDER.S142126
5. Mazoteras-Pardo V et al. The QardioArm Blood Pressure App for Self-Measurement in an Obese Population: Validation Study Using the ESH-IP Revision 2010. *JMIR Mhealth Uhealth*. 2018;6(10):e11632. doi:10.2196/11632
6. Mazoteras-Pardo V et al. QardioArm Upper Arm Blood Pressure Monitor Against Omron M3 in Patients With Chronic Kidney Disease: A Validation Study According to the ESH-IP Revision 2010. *J Med Internet Res*. 2019;21(12):e14686. doi:10.2196/14686
7. Mazoteras-Pardo V et al. QardioArm Blood Pressure Monitoring in a Population With Type 2 Diabetes: Validation Study. *J Med Internet Res*. 2020;22(7):e19781. doi:10.2196/19781
8. Bluetooth SIG, Blood Pressure Profile 1.1.1 and Blood Pressure Service 1.1.1; IEEE 11073-20601 SFLOAT.

All documents: kuardio.com/regulatory.html.

Appendix: every 0x2A35 packet of the 28 September 2026 session

Time (UTC)	Bytes	Decoded by Kuardio
11:27:54	`00 0B 00 00 00 00 00`	sys=11.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:55	`00 13 00 00 00 00 00`	sys=19.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:56	`00 18 00 00 00 00 00`	sys=24.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:56	`00 1C 00 00 00 00 00`	sys=28.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:57	`00 21 00 00 00 00 00`	sys=33.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:57	`00 25 00 00 00 00 00`	sys=37.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:58	`00 29 00 00 00 00 00`	sys=41.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:59	`00 2D 00 00 00 00 00`	sys=45.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:27:59	`00 32 00 00 00 00 00`	sys=50.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:00	`00 37 00 00 00 00 00`	sys=55.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:00	`00 3B 00 00 00 00 00`	sys=59.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:01	`00 40 00 00 00 00 00`	sys=64.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:02	`00 46 00 00 00 00 00`	sys=70.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:02	`00 4B 00 00 00 00 00`	sys=75.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:03	`00 51 00 00 00 00 00`	sys=81.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:03	`00 57 00 00 00 00 00`	sys=87.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:04	`00 5E 00 00 00 00 00`	sys=94.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg

11:28:05	`00 65 00 00 00 00 00`	sys=101.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:05	`04 FF 07 01 00 03 00 00 00`	sys=nil dia=1.0 map=3.0 pulse=0.0 status=none flags=0x04 unit=mmHg
11:28:16	`00 09 00 00 00 00 00`	sys=9.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:17	`00 11 00 00 00 00 00`	sys=17.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:18	`00 18 00 00 00 00 00`	sys=24.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:18	`00 1C 00 00 00 00 00`	sys=28.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:19	`00 20 00 00 00 00 00`	sys=32.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:19	`00 25 00 00 00 00 00`	sys=37.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:20	`00 29 00 00 00 00 00`	sys=41.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:21	`00 2D 00 00 00 00 00`	sys=45.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:21	`00 32 00 00 00 00 00`	sys=50.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:22	`00 37 00 00 00 00 00`	sys=55.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:22	`00 3C 00 00 00 00 00`	sys=60.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:23	`00 42 00 00 00 00 00`	sys=66.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:24	`00 47 00 00 00 00 00`	sys=71.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:24	`00 4D 00 00 00 00 00`	sys=77.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:25	`00 53 00 00 00 00 00`	sys=83.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:25	`00 59 00 00 00 00 00`	sys=89.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:26	`00 60 00 00 00 00 00`	sys=96.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg

11:28:27	`00 67 00 00 00 00 00`	sys=103.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:27	`00 6F 00 00 00 00 00`	sys=111.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:28	`00 76 00 00 00 00 00`	sys=118.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:28	`00 7E 00 00 00 00 00`	sys=126.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:29	`00 86 00 00 00 00 00`	sys=134.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:30	`00 8F 00 00 00 00 00`	sys=143.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:30	`00 97 00 00 00 00 00`	sys=151.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:31	`00 A0 00 00 00 00 00`	sys=160.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:31	`00 A5 00 00 00 00 00`	sys=165.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:32	`00 A1 00 00 00 00 00`	sys=161.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:33	`00 9B 00 00 00 00 00`	sys=155.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:33	`00 98 00 00 00 00 00`	sys=152.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:33	`00 96 00 00 00 00 00`	sys=150.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:34	`00 93 00 00 00 00 00`	sys=147.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:34	`00 8E 00 00 00 00 00`	sys=142.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:35	`00 8A 00 00 00 00 00`	sys=138.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:35	`00 85 00 00 00 00 00`	sys=133.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:36	`00 80 00 00 00 00 00`	sys=128.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:37	`00 7C 00 00 00 00 00`	sys=124.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg

11:28:37	00 78 00 00 00 00 00	sys=120.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:38	00 74 00 00 00 00 00	sys=116.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:38	00 70 00 00 00 00 00	sys=112.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:39	00 6C 00 00 00 00 00	sys=108.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:40	00 69 00 00 00 00 00	sys=105.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:40	00 66 00 00 00 00 00	sys=102.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:41	00 62 00 00 00 00 00	sys=98.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:41	00 5F 00 00 00 00 00	sys=95.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:42	00 5D 00 00 00 00 00	sys=93.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:43	00 5A 00 00 00 00 00	sys=90.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:43	00 57 00 00 00 00 00	sys=87.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:44	00 54 00 00 00 00 00	sys=84.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:44	00 51 00 00 00 00 00	sys=81.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:45	00 4F 00 00 00 00 00	sys=79.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:45	14 90 00 5F 00 6F 00 60 00 00 00	sys=144.0 dia=95.0 map=111.0 pulse=96.0 status=none flags=0x14 unit=mmHg
11:28:45	14 90 00 5F 00 6F 00 60 00 00 00	sys=144.0 dia=95.0 map=111.0 pulse=96.0 status=none flags=0x14 unit=mmHg
11:28:55	00 07 00 00 00 00 00	sys=7.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:56	00 0F 00 00 00 00 00	sys=15.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:56	00 16 00 00 00 00 00	sys=22.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg

11:28:57	`00 1A 00 00 00 00 00`	sys=26.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:58	`00 1E 00 00 00 00 00`	sys=30.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:28:58	`00 22 00 00 00 00 00`	sys=34.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:29:05	`00 07 00 00 00 00 00`	sys=7.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:29:06	`00 0F 00 00 00 00 00`	sys=15.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:29:06	`00 16 00 00 00 00 00`	sys=22.0 dia=0.0 map=0.0 pulse=nil status=none flags=0x00 unit=mmHg
11:29:07	`14 FF 07 01 00 03 00 00 00 00 00`	sys=nil dia=1.0 map=3.0 pulse=0.0 status=none flags=0x14 unit=mmHg